Cameron Emery

Augusta, Maine 04330 • 2074231006 • cemerybusiness@gmail.com • **WWW**: Bold Profile

Summary

Embedded systems engineer specializing in real-time firmware, FPGA-based hardware acceleration, and multi-MCU architectures for safety-critical automotive applications. Experienced in hardware/firmware co-design across bare-metal, RTOS, and embedded Linux platforms, with deep expertise in automotive protocols (CAN, CAN-FD, LIN, SENT) and signal integrity analysis. Proven track record building distributed control systems and HMI platforms with deterministic performance and rigorous root-cause debugging.

Top Skills

- Languages: C, C++, Embedded C, Python, Verilog/SystemVerilog, Assembly, TypeScript
- Protocols: CAN, CAN-FD, LIN, SENT, UART, SPI, I^2C
- Embedded Linux: Yocto / OpenEmbedded, Qt/QML, systemd
- Microcontrollers / Platforms: PIC32MK, dsPIC33, Cortex-M, i.MX8 / i.MX93, RP2040, STM32
- Tools: MPLAB, Keil, oscilloscopes, logic analyzers, JTAG, Git
- Hardware: PCB design, schematic capture, signal integrity analysis, hardware-in-the-loop validation

Experience

Embedded Systems Engineer, 08/2025 to Current
Electronic Mobility Controls – Augusta, ME
Embedded firmware and systems work on automotive accessibility platforms — drive-by-wire shifters, throttle/brake interfaces, and servo-actuated gear controls across multi-MCU architectures (PIC32MK, dsPIC33, Cortex-M, i.MX8/i.MX93).

- Solved a shifter integration problem within the first months on the job by adapting an existing CAN/LIN-based control system to operate over SENT. Designed a custom PCB built around OEM-first operation, asserting control only when the accessibility system requires it — preserving native vehicle behavior by default.
- Develop and debug real-time C/C++ firmware for safety-critical drive-by-wire subsystems; validate against CAN, CAN-FD, LIN, and SENT using oscilloscopes and logic analyzers.
- Lead root-cause investigations on field issues spanning firmware timing, CRC validation, analog signal conditioning, and relay-based switching.
- Improved responsiveness and maintainability of Linux-based HMI (Qt/QML) used in production vehicles.
- Work directly with mechanical, electrical, and production teams to move prototypes into reliable production hardware.

Founder / Software Engineer, 01/2024 to 10/2025
Noctur.io – Boise, Idaho, United States

- Independent SaaS venture building web applications for creators and small businesses.
- Designed and built two full-stack applications end-to-end (architecture, APIs, database, frontend, deployment).
- Implemented RESTful and GraphQL APIs, OAuth 2.0 authentication, and CI/CD deployment pipelines.
- Stack: Laravel, React, PostgreSQL, Django

Websites, Portfolios, Profiles

- www.linkedin.com/in/cameronemery67